Roll No.....

ODD END SEMESTER EXAMINATION DECEMBER - 2025

Name of the Program: **M.Tech** *Semester:* **I**

Name of the Course: **Advanced Power Electronics** *Course Code:* **MEV 102**

Time: 3 hours Maximum Marks: 100

Note:-

(i) All questions are compulsory.

(ii) Answer two sub questions among a, b & c in each main question

(iii) Total marks in each main question are twenty.

(iv) Each question carries 10 marks

Q1	(20 marks)	CO 1
a)	Draw the switching characteristics of power MOSFETs. Define turn- ON delay time, rise time, turn – ON time, turn- OFF delay time, fall time and turn- OFF time.	
b)	Draw the V-I characteristics of a thyristor and explain different operating regions. What is the effect of Gate current on the V-I characteristics of a thyristor?	
c)	Explain the operation of IGBT with the help of neat structural diagram and suitable wave forms.	
Q2	(20 marks)	CO 2
a)	Explain the various modes of operation of Boost DC-DC converter with necessary waveforms.	
b)	Discuss operation of Flyback converter. Draw its circuit diagram and waveforms	
c)	A boost regulator has an input voltage of 5V. The average output voltage is 15V (5) and average load current is 0.5A. The switching frequency is 25kHz.IfL=150uH and C-220uE, determine a) the duty cycle, b) the ripple current c) ripple voltage	
Q3	(20 marks)	CO 3
a)	With the help of a circuit diagram and waveforms, explain the operation of a single-phase half-wave diode rectifier with a R-L load. Derive the expression for average and rms volage.	
b)	A single-phase full-wave bridge diode rectifier is fed from a 230V, 50Hz supply. The load is resistive load 0f R=10ohm. Calculate the following for ideal diodes: (i) Average DC output voltage (Vdc) (ii) RMS DC output voltage (Vrms) (iii) Average DC output current (Idc) (iv) RMS DC output current (Irms) (v) Rectification efficiency	
c)	With the help of a neat circuit diagram and waveforms, explain the operation of 3-phase bridge inverter with R load.	
Q4	(20 marks)	CO 4
a)	Draw the circuit diagram of a single-phase AC voltage controller and explain the working with waveform. Also derive the expression for RMS output voltage, RMS current and average power delivered to the load.	
b)	Explain with the help of a neat circuit diagram and waveforms operation of a three-phase bidirectional delta connected controller.	
c)	A single-phase ac voltage controller has a resistive load of R= 50 ohm and the input voltage is Vs = 150 V 50 Hz. The delay angle of thyristor Tl and T2 is $\pi/2$. Determine (a) The rms value of output voltage Vo (b) the input PF (c) the average thyristor current and d) the RMS value of thyristor current	
Q5	(20 marks)	CO 5
a)	Describe the basic principle of operation of a single phase to single phase bridge type cycloconverter for continuous conduction.	
b)	With relevant waveforms explain the operation of 3 phase to three phase cycloconverter.	
c)	A three pulse cycloconverter has output voltage of 250 to feed single phase load at 50A. if the load p.f is unity, find the input power per phase and power factor.	